Assessing Carryover in the Analysis Model: Standard Methodology and Application to pmsimstats

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2026-03-22

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1 The Question

When simulating clinical trial data with carryover effects in the data-generating process (DGP), what is the proper way to handle carryover in the analysis model? The pmsimstats framework currently offers two options – a continuous drug indicator (`Dbc`) that bakes carryover into the treatment variable, and a separate `tsd` covariate – but the choice between them (or whether to include carryover at all) has not been systematically evaluated.

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2 The 2x2 Factorial Framework

The standard simulation framework treats the DGP and analysis model as a factorial design with two factors:

	Analysis ignores carryover	Analysis models carryover
DGP: no carryover	Baseline (Type I error check)	Cost of unnecessary adjustment
DGP: carryover present	Cost of ignoring carryover	Correct specification

Data are generated under each DGP condition, analyzed under each analysis model, and the following metrics are reported for all four cells:

- Type I error (rejection rate when $c_{bm} = 0$)
- Power (rejection rate when $c_{bm} > 0$)
- Bias of the treatment effect estimate
- Coverage of the 95% confidence interval

The diagonal cells (matched DGP and analysis) should perform well. The off-diagonal cells reveal the consequences of model misspecification: inflated Type I error when carryover is ignored, or unnecessary power loss when carryover is modeled but absent.

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3 What the Literature Says

3.1 The definitive reference

Senn (2002), *Cross-over Trials in Clinical Research*, 2nd edition, provides the authoritative treatment. The key conclusions for simulation design:

1. **The two-stage procedure is invalid.** The older approach – test for carryover first, then decide whether to include it in the treatment model – was shown to be statistically unsound. The carryover test has low power (crossover designs are not designed to detect carryover; they are designed to detect treatment effects). Conditioning the treatment analysis on the outcome of the carryover test inflates the Type I error of the treatment test beyond the nominal level.
2. **The modern consensus:** If carryover is scientifically plausible, either (a) design the trial to eliminate it via a washout period of sufficient length, or (b) always include carryover in the model and accept the associated power loss. Do not test-then-decide.
3. **For simulation studies:** Generate data under a range of carryover magnitudes (including zero), analyze under models that include and exclude the carryover term, and report operating characteristics for each combination.

3.2 Supporting references

- Jones & Kenward (2014), *Design and Analysis of Cross-Over Trials*, 3rd edition. Extends Senn's framework to higher-order crossover designs and provides simulation templates for evaluating carryover model specifications.
- Zucker et al. (1997), 'Combining single patient (N-of-1) trials to estimate population treatment effects and its application to crossover designs.' Establishes the mixed-effects framework for aggregated N-of-1 trials, which is the basis of the pmsimstats analysis model.
- Araujo et al. (2016), 'Understanding variation in sets of N-of-1 trials.' Addresses heterogeneity in treatment effects across participants, which is precisely the biomarker-treatment interaction that pmsimstats is designed to detect.

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4 How This Maps to pmsimstats

4.1 Carryover in the DGP

The pmsimstats DGP introduces carryover through the mean structure of the biological response (BR) component. During off-drug periods, the BR mean at timepoint t is augmented by the decayed effect from the previous timepoint:

$$\mu_{BR}(t) = \text{base}(t) + \mu_{BR}(t-1) \cdot (1/2)^{t_{sd}/t_{1/2}}$$

where t_{sd} is the time since drug discontinuation and $t_{1/2}$ is the carryover half-life. This is an exponential decay model: the residual drug effect halves every $t_{1/2}$ weeks.

Additionally, the BM-BR correlation decays during off-drug periods at a rate tied to the same half-life:

$$\text{Cor}(bm, br_t) = c_{bm} \cdot e^{-\lambda_{cor} \cdot t_{sd}}, \quad \lambda_{cor} = \ln(2)/t_{1/2}$$

4.2 Carryover in the analysis model

The current codebase offers two approaches:

Approach A: Continuous drug indicator (`Dbc`). The binary on/off-drug indicator is replaced with a continuous version:

- On drug: $Dbc = 1$
- Off drug: $Dbc = (1/2)^{s \cdot t_{sd}/t_{1/2}}$

The model formula becomes: `Sx ~ bm + t + Dbc + bm:Dbc`, with the interaction `bm:Dbc` as the test statistic.

This approach conflates treatment status and carryover into a single variable. The `bm:Dbc` coefficient captures both the treatment-biomarker interaction and any residual carryover-biomarker interaction.

Whether this is a feature or a bug depends on the scientific question: if the goal is to detect 'does the biomarker predict who benefits from drug exposure (including residual exposure),' then `Dbc` is appropriate. If the goal is to detect 'does the biomarker predict the on-drug treatment effect, adjusted for carryover,' then a separate carryover term is needed.

Approach B: Separate carryover covariate (`tsd`). A linear time-since-discontinuation term is added to the model:

```
Sx ~ bm + t + Db + bm:Db + tsd
```

where `Db` is binary (on/off drug) and `tsd` absorbs the carryover effect. The interaction `bm:Db` then estimates the biomarker-treatment interaction after adjusting for carryover. This is closer to the standard crossover trial approach but uses a linear carryover effect, which is a mild misspecification when the DGP uses exponential decay.

Approach C: Ignore carryover. The model uses only the binary drug indicator:

```
Sx ~ bm + t + Db + bm:Db
```

Off-drug observations with residual drug effect are treated as if no drug were present. This biases the treatment contrast toward zero (carryover makes off-drug periods look more like on-drug periods), reducing power but potentially inflating Type I error if the bias is systematic.

4.3 The non-standard element

The `Dbc` parameterization (Approach A) is creative but non-standard in the crossover trial literature. Standard practice separates the treatment indicator from the carryover adjustment. The `Dbc` approach would require explicit justification in a methods paper, ideally supported by the factorial simulation comparison described below.

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5 Proposed Simulation Study

A rigorous evaluation would compare three analysis models across the carryover DGP sweep.

5.1 Design

DGP factor: $t_{1/2} \in \{0, 0.25, 0.5, 1.0, 2.0\}$ weeks

Analysis factor: Three models:

1. **Ignore:** `Sx ~ bm + t + Db + bm:Db` (binary drug indicator, no carryover term)
2. **Separate:** `Sx ~ bm + t + Db + bm:Db + tsd` (binary drug indicator + linear carryover covariate)
3. **Continuous:** `Sx ~ bm + t + Dbc + bm:Dbc` (exponential-decay drug indicator, current default)

Fixed parameters: $N = 70$, $\rho = 0.7$, all four trial designs.

Biomarker correlation: $c_{bm} \in \{0, 0.3, 0.45\}$ (null, moderate, strong).

Replicates: 1,000 per cell.

5.2 Metrics

For each of the $5 \times 3 \times 3 \times 4 = 180$ cells:

- **Type I error** (at $c_{bm} = 0$): should be ≤ 0.05
- **Power** (at $c_{bm} > 0$): higher is better
- **Bias:** $E[\hat{\beta}] - \beta_{true}$
- **RMSE:** $\sqrt{E[(\hat{\beta} - \beta_{true})^2]}$
- **Coverage:** proportion of 95% CIs containing β_{true}

5.3 Expected findings

Based on the literature and the DGP structure:

- **Ignore (model 1)** will have correct Type I error at $t_{1/2} = 0$ but may have inflated Type I error at higher carryover values (residual drug effect creates a systematic correlation between biomarker and off-drug response that the model attributes to noise). Power will decline with increasing carryover.
- **Separate (model 2)** will have correct Type I error across all carryover values (the `tsd` term absorbs the carryover signal). Power will be slightly lower than model 1 at $t_{1/2} = 0$ (cost of the extra parameter) but higher at $t_{1/2} > 0$ (correct adjustment recovers treatment contrast). Mild bias possible due to linear vs exponential misspecification.
- **Continuous (model 3)** will have correct Type I error if the `Dbc` decay rate matches the DGP. Power behavior depends on whether the conflation of treatment and carryover helps or hurts: it may outperform model 2 when the carryover half-life is correctly specified, but underperform when misspecified.

5.4 Infrastructure

The existing codebase supports all three models:

- Model 1: `op$carryover_t1half = 0, op$simplecarryover = FALSE`
- Model 2: `op$carryover_t1half = 0, op$simplecarryover = TRUE`
- Model 3: `op$carryover_t1half = [matching DGP value]`

The simulation grid would need to cross the DGP carryover parameter with the analysis model choice. This requires a modification to `generateSimulatedResults()` (or the 2025 equivalent) to accept multiple analysis parameter sets per DGP configuration, which is already supported via the `analysisparams` argument.

6 Recommendation

The factorial comparison described in Section 5 should be run before finalizing the manuscript. It addresses two questions that reviewers will ask:

1. **Does the `Dbc` parameterization control Type I error?** If it does not, the power estimates reported in the paper are unreliable.
2. **Is the power gain from modeling carryover worth the complexity?** If the 'ignore' model performs nearly as well as the carryover-aware models, the simpler approach is preferable.

The simulation infrastructure already exists. The primary effort is configuring the parameter grid and collecting the results into a comparison table. Estimated effort: 4–6 hours of computation (parallelizable) plus 2–3 hours of analysis and writing.

Rendered on 2026-03-25 at 19:00 PDT.

Source: ~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/docs/10-carryover-analysis-model-assessment.tex